The Neptunian Harmonic Lattice
Orbital Architecture, Phase Debt Distribution, and Boundary Closure in the Neptune System

Annex NEP to Harmonic Orbital Mechanics

Harmonic Structures in Natural Systems

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"*There is geometry in the humming of the strings,*
there is music in the spacing of the spheres."
— Pythagoras

Abstract

This paper derives the complete orbital architecture and boundary mechanics of the Neptunian system from first principles, using discrete frequency-phase resonance mapping within the Unified Harmonic Shells (UHS) framework. By eliminating continuous gravitational parameters (G) and arbitrary mass terms, the configuration of Neptune's fifteen satellites and three primary ring boundaries is formalized using a global lattice resolution of k = 41, anchored to a locked macro-orbital baseline of 164 + 11/14.

The analysis demonstrates that the spatial positions and orbital periods of the inner prograde sequence and outer irregular cluster are locked features of a discrete, self-interfering resonator. Mapped through the velocity-frequency transformation operator $\Phi(r) = r^{-3/2}$, ring boundaries are integrated directly into the global phase debt budget. The system resolves to a non-vanishing residue of ε = −36,219.40 s (≈ −10.061 hours), the measurable signature of an active planetary oscillator rather than a closure failure. The stable latitude of the Great Dark Spot is derived as a direct function of a field inversion where Triton's retrograde vector dictates the directional assignment of the 1/3 torque invariant. Clear, non-adjustable boundary constraints are established for direct empirical falsification.

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1. Introduction and System Parameters

1.1 Observational Framework

All observations are made from Earth. Earth's orbital period is therefore the fundamental baseline clock against which all other periods are compared. No assumptions about universal constants, masses, or forces are required beyond the raw observational record. All raw temporal values and spatial measurements are sourced directly from NASA JPL Horizons ephemerides. No rounding or approximation is applied at the data import stage.

1.2 Framework Foundation

This paper applies the following foundational operators and invariants, each fully derived in their respective source papers. Foundational justifications — including the derivation of the n = 4k admissibility rule, the 11/14 phase invariant, and the nearest-integer lattice selection criterion — reside in the cited source papers and are not re-derived here. No derivation is repeated in this annex.

Operator / Law	Core Result	Source
Phi-Operator ($\Lambda^2 \cdot \Phi$)	$f(r) = \Lambda^2 / r^{3/2}$	Johnson (2025a), SemanticDrift
Law of Admissibility	n = 4k; R = 4	Johnson (2025b), SemanticDrift
Origami Principle	C = (22/7)D	Johnson (2025c), SemanticDrift
Unified Harmonic Shells (UHS)	$r_n = c / (n \cdot f_0)$	Johnson (2025d), SemanticShift
Taylor Expansion / 1/6 Torsion	Torsion shell at n = 3	Johnson (2025e), SemanticShift
Harmonic Distance Law	r = 1/f (outer chamber)	Johnson (2025f), SemanticShift
Annex JUP / Annex SAT	1/3 invariant (atmospheric)	Johnson (2025g,h), SemanticShift

Note on the Origami Principle: All circular closure operations in this paper use the rational fraction 22/7, derived from discrete folding geometry (Johnson, 2025c). This is not a claim that 22/7 equals π, nor a rejection of π within continuous analysis. Discrete folding produces exact rational closure; continuous idealization produces transcendental approximation. These are different frameworks applied to different domains. Readers are directed to The Origami Principle for the full derivation.

Note on the 1/3 invariant: The 1/3 rational fold as a scale-invariant atmospheric anchor is established in Annex JUP (Great Red Spot: $R_J \times 1/3 \approx 23{,}830$ km, observed ≈ 24,000 km, 0.7%) and Annex SAT (Great White Spot: $R_S \times 1/3 \approx 20{,}089$ km, observed ≈ 20,000 km, 0.4%). Annex NEP extends this invariant to Neptune's Great Dark Spot.

1.3 Neptune System Parameters

All values sourced from NASA JPL Horizons ephemerides.

Parameter	Value
Equatorial Radius (R_N)	24,622 km
Sidereal Orbit (T_N)	60,189d 05h 45m 36s = 5,200,350,336 s
Synodic Period (S_N)	367d 11h 45m 36s = 31,751,136 s
Core Rotation (t_N)	0d 16h 06m 36s = 57,996 s

1.4 Satellite Observed Periods (NASA JPL Horizons)

Negative periods denote retrograde orbits.

Inner Regular Satellites (Prograde)

Satellite	Raw Period	Total Seconds
Naiad (t_1)	0d 07h 03m 55.79s	25,195.79 s
Thalassa (t_2)	0d 07h 28m 32.25s	26,912.25 s
Despina (t_3)	0d 08h 01m 54.24s	28,914.24 s
Galatea (t_4)	0d 10h 17m 23.51s	37,043.51 s
Larissa (t_5)	0d 13h 18m 42.05s	47,922.05 s
Hippocamp (t_6)	0d 22h 48m 33.70s	82,113.70 s
Proteus (t_7)	1d 02h 56m 08.00s	96,968.00 s

Main Satellite (Retrograde)

Satellite	Raw Period	Total Seconds
Triton (t_8)	−5d 21h 02m 37.00s	−507,757.00 s

Outer Irregular Satellites

Satellite	Raw Period	Total Seconds
Nereid (t_9)	360d 03h 21m 36.00s	31,116,096.00 s
Halimede (t_{10})	−1,879d 01h 55m 12.00s	−162,350,112.00 s
Sao (t_{11})	2,912d 17h 16m 48.00s	251,659,008.00 s
S/2002 N 5 (t_{12})	3,156d 13h 20m 38.00s	272,726,438.00 s
Laomedeia (t_{13})	3,171d 07h 55m 12.00s	274,002,912.00 s
Psamathe (t_{14})	−9,149d 12h 20m 10.00s	−790,520,410.00 s
Neso (t_{15})	−9,794d 16h 55m 12.00s	−846,261,312.00 s
S/2021 N 1 (t_{16})	−10,036d 15h 37m 26.00s	−867,209,846.00 s

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2. Establishing the Neptune-Earth Macro-Orbital Baseline

2.1 Raw Observational Data

All raw temporal values are imported directly from NASA JPL ephemerides and converted to pure integer seconds. No rounding or approximation is applied at this stage.

Parameter	Raw Observation	Total Seconds
Earth Sidereal Orbit (T_E)	365.25636 days	31,558,150 s
Neptune Sidereal Orbit (T_N)	60,189d 05h 45m 36s	5,200,350,336 s

Conversion shown explicitly for Neptune:

T_N = 60,189 days + 5 hours + 45 minutes + 36 seconds
= (60,189 × 86,400) + (5 × 3,600) + (45 × 60) + 36
= 5,200,329,600 + 18,000 + 2,700 + 36
= **5,200,350,336 seconds**

Earth's orbital period taken as the standard sidereal year:

$$T_E = 365.25636 \times 86{,}400 = 31{,}558{,}150 \text{ s}$$

2.2 The Raw Orbital Frequency Ratio

$$T_N / T_E = 5{,}200{,}350{,}336 / 31{,}558{,}150 = \mathbf{164.7860137...}$$

This is the number of Earth orbits per Neptune orbit. The value is non-integer, indicating the presence of phase debt at the macro-orbital interface.

2.3 Applying the Law of Admissibility (n = 4k)

The Law of Admissibility (Johnson, 2025b) states that a system achieves coherence when its primary frequency boundary locks onto an integer node n satisfying n = 4k, where k is the system's memory depth. The derivation of this rule — including why the nearest multiple of 4 is selected rather than the nearest integer — is provided in full in Johnson (2025b). Nearest multiples of 4 to the raw ratio:

$|164.7860137 - 164| = 0.7860137$ ← minimum
$|164.7860137 - 168| = 3.2139863$

The closest is n = 164. Solving for k:

$$4k = 164 \rightarrow k = 164 / 4 = \mathbf{41}$$

The Neptune-Earth macro-orbital interface has a memory depth of k = 41.

2.4 Isolating the Phase Debt

$$F = 164.7860137... - 164 = \mathbf{0.7860137...}$$

2.5 Converting Phase Debt to a Rational Fraction

The phase debt 0.7860137... is identified as a rational fraction derived from the Origami Principle (Johnson, 2025c) filtered through the Law of Admissibility (R = 4). The derivation of why the Origami Principle generates this particular rational is provided in full in Johnson (2025c).

$$(22/7) \times (1/4) = 22/28 = \mathbf{11/14} = 0.785714...$$

Observed phase debt: 0.7860137...
Rational node (11/14): 0.785714...
Micro-residual: 0.0002997...

In exact rational form:

$$T_N/T_E = (164 \times 14 + 11) / 14 = 2{,}307/14 = \mathbf{164.785714...}$$

2.6 Section 2 Summary

Parameter	Value
Raw ratio T_N/T_E	164.7860137...
Nearest admissible node n = 4k	164
Memory depth k	41
Phase debt F	0.7860137...
Geometric invariant fraction	11/14 = 0.785714...
Locked macro-ratio	164 + 11/14 = 2,307/14
Micro-residual	0.0002997...

The system-specific invariant for Neptune is established: k = 41, lattice resolution ε_N = 1/41, macro-orbital lock at 2,307/14.

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3. The Neptunian Cavity: Satellite and Ring Mapping

3.1 Inner Prograde Cavity: Node Assignment Protocol

The inner prograde cavity is anchored to Proteus (T_P = 96,968.00 s), the outermost and most massive of the inner regular satellites. All inner satellite nodes are assigned by finding the nearest integer n within the k = 41 lattice such that (41/n) × T_P minimizes the residual against the observed period. The requirement for discrete lattice occupancy — and therefore nearest-integer selection — is derived in the UHS framework (Johnson, 2025d).

$$n = \mathrm{round}(41 \times T_P / T_{obs}) \quad T_{derived} = (41/n) \times T_P$$

3.2 Node Assignment Table

Satellite	T_{obs} (s)	Calculation (nearest integer)	n
Naiad	25,195.79	41 × 96,968 / 25,195.79 = 157.83	158
Thalassa	26,912.25	41 × 96,968 / 26,912.25 = 147.74	148
Despina	28,914.24	41 × 96,968 / 28,914.24 = 137.52	138
Galatea	37,043.51	41 × 96,968 / 37,043.51 = 107.34	107
Larissa	47,922.05	41 × 96,968 / 47,922.05 = 82.95	83
Hippocamp	82,113.70	41 × 96,968 / 82,113.70 = 48.43	48
Proteus	96,968.00	Anchor node	41

3.3 Inner Cavity Closure Matrix

Satellite	Node	Derived Period (s)	Observed (s)	Residual (s)	Var. (%)
Naiad	158/41	41/158 × 96,968 = 25,162.63	25,195.79	+33.16	0.13
Thalassa	148/41	41/148 × 96,968 = 26,862.92	26,912.25	+49.33	0.18

Satellite	Node	Derived Period (s)	Observed (s)	Residual (s)	Var. (%)
Despina	138/41	41/138 × 96,968 = 28,811.25	28,914.24	+102.99	0.36
Galatea	107/41	41/107 × 96,968 = 37,155.96	37,043.51	−112.45	0.30
Larissa	83/41	41/83 × 96,968 = 47,900.29	47,922.05	+21.76	0.05
Hippocamp	48/41	41/48 × 96,968 = 82,826.83	82,113.70	−713.13	0.87
Proteus	41/41	41/41 × 96,968 = 96,968.00	96,968.00	0.00	0.00
Mean abs. error					**0.27**

Note: Hippocamp carries the largest individual residual (0.87%), consistent with its status as a recently identified satellite believed to be a fragment of Proteus. This residual is noted and not suppressed.

3.4 Triton: The Outer Cavity Reference Boundary

Triton orbits Neptune retrogradely ($T_T = -507{,}757.00$ s). It functions as the reference boundary condition for the outer irregular cavity, analogous to Proteus in the inner cavity. Its retrograde trajectory imposes a directional phase reversal across the outer field. Under the chosen coordinate system, Triton's residual is identically zero because it defines the reference frame against which outer cavity nodes are measured, not because it was set to zero arbitrarily.

3.4.1 Outer Irregular Cavity: Node Assignment

$T_{derived} = (n/41) \times |T_T|$ (negative sign preserved for retrograde bodies)

3.4.2 Outer Cavity Closure Matrix

Satellite	Node	Derived Period (s)	Observed (s)	Residual (s)	Var. (%)
Nereid	2513/41	31,122,654.80	31,116,096.00	−6,558.80	0.021
Halimede	−13109/41	−162,345,699.29	−162,350,112.00	−4,412.71	0.003
Sao	20321/41	251,660,115.34	251,659,008.00	−1,107.34	0.0004
S/2002 N 5	22022/41	272,725,572.54	272,726,438.00	+865.46	0.0003
Laomedeia	22125/41	274,001,155.49	274,002,912.00	+1,756.51	0.0006
Psamathe	−63832/41	−790,517,764.49	−790,520,410.00	−2,645.51	0.0003

Satellite	Node	Derived Period (s)	Observed (s)	Residual (s)	Var. (%)
Neso	−68331/41	−846,236,419.85	−846,261,312.00	−24,892.15	0.003
S/2021 N 1	−70025/41	−867,212,311.95	−867,209,846.00	+2,465.95	0.0003
Mean abs. error					**0.004**

3.5 Ring Cavity Boundary Mapping and Resonance Verification

The ring system is mapped using the same k = 41 resolution framework. A dual-anchor reference frame is used: the innermost Galle ring anchors directly to the planetary radius, whereas the Le Verrier and Adams rings occupy the sub-Proteus zone and reference the inner satellite boundary (r_P = 117,647 km, NASA JPL).

3.5.1 The Galle Ring (Planetary Radius Anchor)

Verification: 41,900 / 24,622 × 41 = 69.77 → nearest integer node n = 70

$$r_{Galle} = 24{,}622 \times (70/41) = \mathbf{42{,}037.56\ km}$$

Observed: 41,900 km Micro-residual: +137.56 km Variance: 0.33%

3.5.2 The Le Verrier Ring

Verification: 53,200 / 117,647 × 82 = 37.07 → node n = 37, denominator 82 = 2 × 41

$$r_{LeVerrier} = 117{,}647 \times (37/82) = \mathbf{53{,}084.62\ km}$$

Observed: 53,200 km Variance: −0.22%

3.5.3 The Adams Ring (Arc Attractor)

Verification: 62,933 / 117,647 × 41 = 21.93 → node n = 22

$$r_{Adams} = 117{,}647 \times (22/41) = \mathbf{63{,}127.68\ km}$$

Observed: 62,933 km Variance: +0.31%

The Adams ring contains stable, localized arc structures (Fraternité, Égalité, Liberté). These arcs are direct empirical evidence of non-axisymmetric phase-locking: matter trapped in discrete angular wave-packets at the outer clamped boundary of the high-frequency envelope.

Ring	Anchor	Rational Node	Derived (km)	Observed (km)	Variance
Galle	R_N	70/41	42,037.56	41,900	+0.33%
Le Verrier	r_P	37/82	53,084.62	53,200	−0.22%
Adams	r_P	22/41	63,127.68	62,933	+0.31%
Mean abs. error					**0.29%**

3.5.4 Ring Spacing Harmonic Ratios

The spacing between ring boundaries independently recovers the system's macro-orbital invariants:

Le Verrier / Galle = 53,200 / 41,900 = 1.2697 ≈ 14/11 = 1.2727 (variance 0.24%)

Adams / Galle = 62,933 / 41,900 = 1.5020 ≈ 3/2 = 1.5000 (variance 0.13%)

The ratio 14/11 is the reciprocal of the macro-orbital geometric invariant 11/14 derived in Section 2. Its appearance in the ring spacing confirms that the same rational structure governs the system from the macro-orbital scale down to the ring envelope. The Adams ring closes at the 3/2 interval from the Galle anchor: the perfect fifth of the harmonic series.

3.5.5 Cross-Layer Verification via the Phi Operator

Normalized ring radii (r/r_p) are passed through $\Phi(r) = r^{-3/2}$ to verify continuity between ring and satellite layers:

Le Verrier: r/r_p = 0.45219 Φ(0.45219) = 3.291 [Despina node: 138/41 ≈ 3.366]

Adams: r/r_p = 0.53493 Φ(0.53493) = 2.556 [Galatea node: 107/41 ≈ 2.610]

These values confirm approximate correspondence between ring boundary positions and the frequency zones of adjacent inner satellites, demonstrating continuity across the ring and satellite layers.

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4. Atmospheric Anchoring: The Great Dark Spot Latitude

4.1 The 1/3 Structural Invariant

In Annex JUP (Johnson, 2025g) and Annex SAT (Johnson, 2025h), the 1/3 rational fold is established as a scale-invariant boundary. When the internal cavity's phase energy reaches this precise geometric node, it projects onto the planet's atmospheric layer as a stable vortex feature. Annex NEP extends this invariant to Neptune.

4.2 Great Dark Spot Latitude Derivation

Applying the 1/3 rational fold to the system-specific memory depth k = 41:

$$\text{Lattice Boundary Node} = k/3 = 41/3 = 13.6$$

$$13.6 / 41 = 1/3 \text{ (exact)}$$

Angular projection onto Neptune's surface (maximum quadrant 90° from R = 4):

$$\text{Unfitted projection: } 90° \times 1/3 = \mathbf{30.000°}$$

Phase debt correction using macro-orbital micro-residual Δ = 0.0002997 from Section 2:

$$\text{Derived Latitude} = 30° - (90° \times 0.0002997) = \mathbf{29.973° \ South}$$

The southern hemisphere assignment results from Triton's retrograde orbital vector. The field inversion coefficient equals 1 in magnitude; Triton's retrograde trajectory is the directional operator that projects the result into the southern hemisphere rather than the northern.

4.3 Verification

Voyager 2 flyby data (1989) recorded Neptune's Great Dark Spot (GDS-89) at 27° to 30° South. The derived 29.973° South falls at the poleward boundary of the observed vortex range, consistent with the 1/3 rational fold node.

Parameter	Derived	Observed	Notes
GDS Latitude	29.973° South	27°–30° South	Poleward boundary of vortex

The 1/3 invariant, confirmed across Jupiter (0.7%) and Saturn (0.4%), holds at the poleward boundary of the Neptunian atmospheric vortex.

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5. Phase Debt and Lattice Closure

5.1 Mathematical Conservation of Phase

Within the UHS framework (Johnson, 2025d), deviations from exact lattice nodes are recognized as structured, non-vanishing phase debt. The system achieves global equilibrium not by erasing this debt, but by conserving it across the boundary layers of the dual-chamber cavity. The conservation condition is:

$$\Sigma\, \Delta\varphi_{\text{inner}} + \Sigma\, \Delta\varphi_{\text{rings}} + \Sigma\, \Delta\varphi_{\text{outer}} + \Delta\varphi_{\text{Triton}} \rightarrow \text{bounded residue } \varepsilon$$

5.2 The Phase Debt Operator

Spatial boundaries are converted to frequency-domain components via the Phi-Operator (Johnson, 2025a): $\Phi(r) = r^{-3/2}$. Frequency residuals ($\Delta\Phi$) are converted to time-domain phase debt in seconds by multiplication with the inner cavity anchor period:

$$\Delta\varphi = \Delta\Phi \times T_{\text{Proteus}} = \Delta\Phi \times 96{,}968 \text{ s}$$

5.3 Ring Cavity Phase Energy

Using normalized radii (r/r_P) relative to r_P = 117,647 km (NASA JPL):

Ring	r / r_P	$\Phi(r/r_P)$	Nearest Node n/41	$\Delta\Phi$	$\Delta\varphi$ (s)
Galle	0.35614	4.70420	193/41 = 4.70731	−0.00311	−301.57
Le Verrier	0.45219	3.28910	135/41 = 3.29268	−0.00358	−347.15
Adams	0.53493	2.55660	105/41 = 2.56097	−0.00437	−423.75
Ring Sum					−1,072.47

5.4 Comprehensive System-Wide Phase Budget

Inner sum derivation (7 satellites):

$$+33.16 + 49.33 + 102.99 - 112.45 + 21.76 - 713.13 + 0.00 = \mathbf{-618.34\ s}$$

Outer sum derivation (8 satellites):

$$-6{,}558.80 - 4{,}412.71 - 1{,}107.34 + 865.46 + 1{,}756.51$$
$$- 2{,}645.51 - 24{,}892.15 + 2{,}465.95 = \mathbf{-34{,}528.59\ s}$$

Layer	Sum (s)
Inner prograde moons (7)	−618.34
Ring boundaries (3)	−1,072.47
Outer irregular moons (8)	−34,528.59
Triton (reference boundary condition)	0.00
Total system residue ε	**−36,219.40 s (≈ −10.061 hours)**

5.5 Geometric Boundary Limits

The non-vanishing system-wide residue (ε = −36,219.40 s) scales proportionally with the local grid metrics of the inner cavity. Rather than matching a continuous velocity vector or an external planetary clock, this residue represents a boundary limit. The unclosed phase debt remains finite and stable, commensurate in scale with the discrete lattice geometry of the inner cavity. Its non-vanishing character is the measurable signature of an active planetary oscillator rather than a closed or stagnant system.

Across the annex series, distinct resolution mechanisms have been identified: Neptune carries a bounded non-zero residue consistent with continuous internal activity; the Pluto-Charon binary distributes its phase debt across orbital, rotational, and barycentric mechanisms summing to ≈ 1/e (Annex PLU, Johnson 2025j); the Uranian system localizes residue into a finite rational defect set while regular lattice seats achieve δ = 0 (Annex URA, Johnson 2025i).

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6. Predictions and Falsification

To ensure the integrity of the k = 41 lattice model, the framework provides clear, non-adjustable parameters explicitly open to empirical verification.

6.1 Atmospheric Vector Lock

The stability of the Great Dark Spot latitude relies entirely on the directional inversion driven by Triton's retrograde orbital vector (k = 41 shear zone boundary). If high-resolution atmospheric tracking demonstrates that the core convective anchoring of the GDS drifts past the strict limits of the k = 41 lattice shear coordinates under long-term observation, the spatial invariant assignment in Section 4 is falsified.

6.2 Ring System Spatial Drift Thresholds

Because the ring boundary tracks are anchored to discrete fractional nodes (193/41, 135/41, and 105/41), their positions cannot undergo continuous, unconstrained secular drift. The framework predicts that any detected variation in ring boundary positions will oscillate around the derived lattice nodes rather than migrating continuously away from them. Secular drift exceeding these micro-residual bounds without reset points would break the global conservation principle and falsify the specific lattice assignment.

6.3 Outer Cavity Metric Verification

The outer irregular moon cluster carries the heaviest component of the system-wide residue ($\Sigma\ \Delta\varphi_{outer} = -34{,}528.59$ s). If orbital tracking of the outermost bodies (e.g., Neso, Psamathe) reveals long-term variations that permanently alter this sum beyond the established $\varepsilon = -10.061$-hour system profile, the framework requires revision at the outer cavity node assignments.

Conclusion

The derivation of the Neptunian cavity confirms that complex multi-body planetary architectures can be evaluated entirely through discrete harmonic constraints. By treating the local system as a conserved phase loop under the Law of Admissibility ($R = 4$), the positions of rings, regular prograde moons, and highly elliptical irregular satellites resolve into an integrated, self-consistent lattice.

The evaluation reveals that an active system does not require a zero-sum phase debt identity. Instead it stabilizes toward a finite, non-vanishing bounded residue (ε), commensurate in scale with the inner cavity's lattice resolution. This residue distinguishes active planetary oscillators from stagnant or binary configurations and provides a measurable, system-specific fingerprint.

Eliminating adjustable parameters forces the framework to stand entirely on its own rigidity, shifting the study of planetary systems away from continuous empirical curve-fitting and into the domain of native harmonic mechanics.

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